UNIT - 1

Need for a database Management System :→

* Earlier Information was stored in the form of spread shut / excel shut, which was not automated. Therefore ease of access was cumbersome & time taking.
* Strategy or decision making was difficult.
* Management of an unautomated data warehouse (data storage) was difficult.
* Every time, when a new problem arises a new application is required.

Database system Applications :→

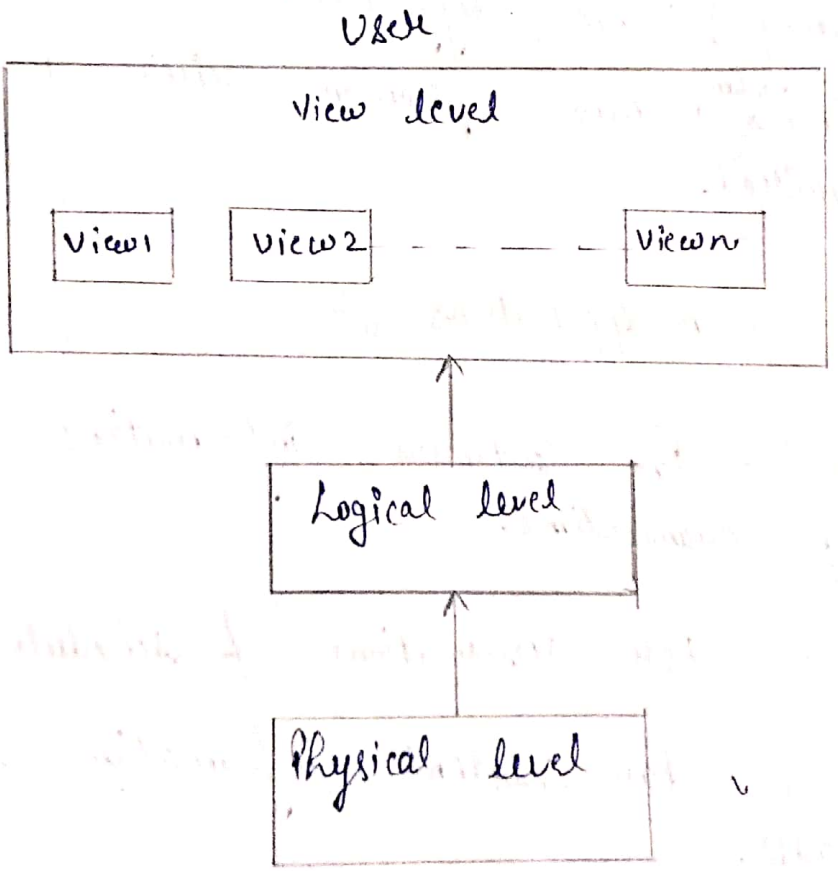
* Banking :→ For customers informations loans, accounts & banking transections.
* Airlines :→ For reservations & schedule informations.
* Universities :→ For student information course registration & grades.
* Credit Card transections :→ For purchases on credit cards & monthly installments.
* Human Resource :→ For information about employees, salary & payroll taxes.

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Purpose of database System :-

* It should remove data redundancy & inconsistancy. (Redundancy → Copying of data).
* Difficulty in accessing data.
* Data isolation.
* Concurrent access enomaly.
* Security Problem.

View of data :-



* Physical Level :- The lowest level of abstraction describes how the data are actually stored. The Physical level describes complex low level data structure in detail.

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Logical level :- The next higher level of abstraction describes what data are stored in the database & what relationship exist among those data. The logical level thus describe the entire data base in terms of a small no. of relatively simple structure

View level :- The highest level of abstraction describe only part of the entire database. The view level of abstraction exist to simplify there interaction with the system.

Instances :- Databases change over time as information is inserted & deleted. The collection of information stored in the database at a particular moment is called an instance of the database.

Schema :- The overall design of the database is called the database schema. The physical schema describes the database design at the physical level. while the logical schema describes the data base design at the logical level.

Physical data independence :- Application programes are said to exhibits physical data independence if they do not depend on the physical schema & thus need not to be rewritten if the physical schema changes.

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Database Languages :→

The language used for creating a database management system is named as structured Query language (SQL)

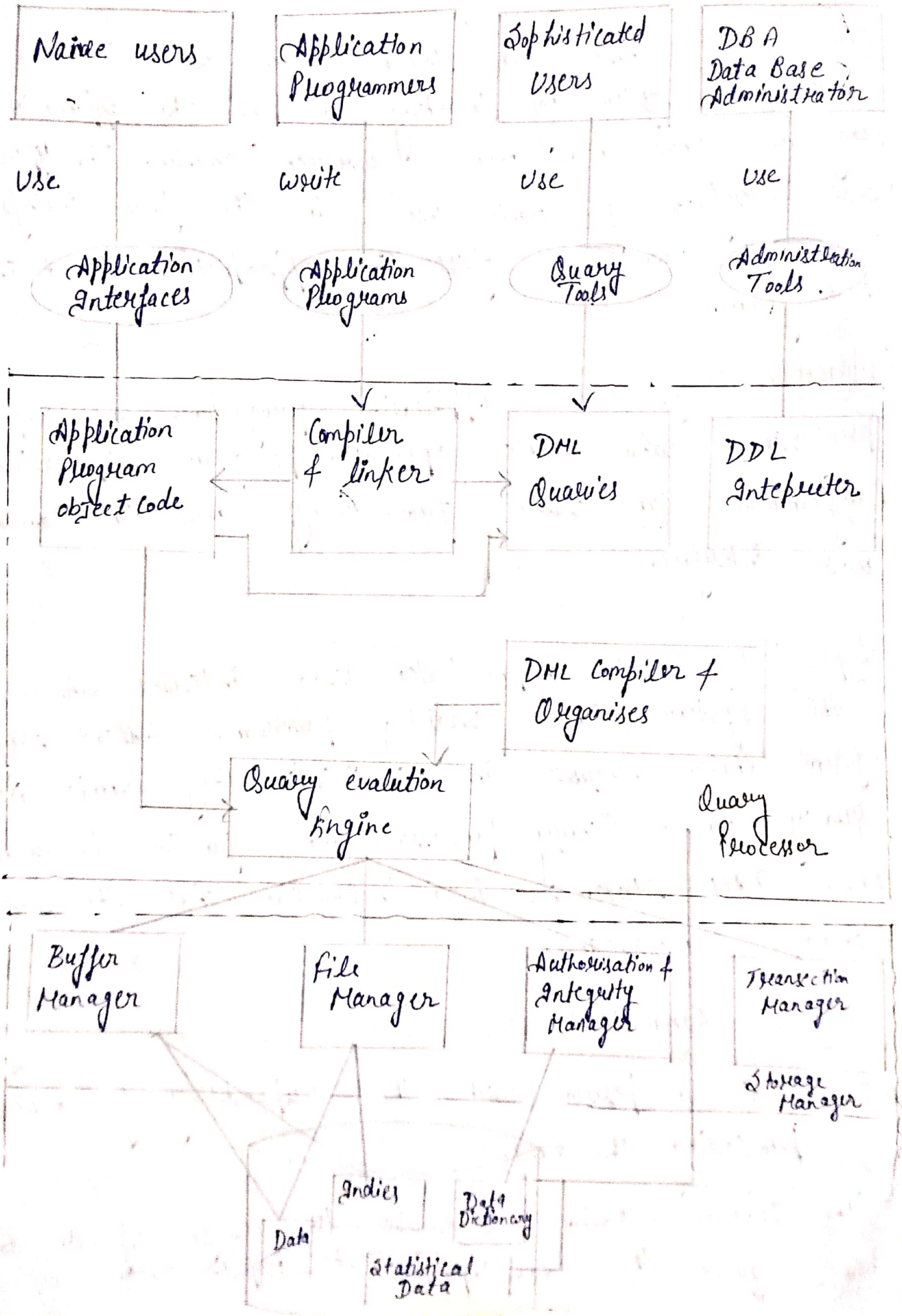
* Data defination Language :→ SQL provides a such (DDL) that allows one to define tables integrity constrains etc.

Data Manuplation language (DML) :→ It is a language that enables users to access or Manuplate data as organised by the appropriate data Model. The Types of access are :→

(i) Retrieval :→ Retrieval of the information from the database.
(ii) Insertion of new information into the D.B.
(iii) Deletion of information from the D.B.
(iv) Modification of information stored in the D.B.

Database Architecture :→

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Database User :→

Naive Users :→ These are simple users who interact with the system by invoking one of the application programs that have been written earlier. The typical user interface for navie user in the form interface. where the user can fill in appropriates field in the form.

Application Programmers :→ Application programmer are computer professional who write application program. Application Programmer can choose from many tools to developed user interface.

Shophisticated users :→ Sophisticated user interact who with the system without writing programs indlead they form their request in a DSL. They submit each query to a Quary processor. whose function is to break down DML statements into instruction that the storage Manager understands.

Database Administrator :→ (DBA)

* DBA is a person with the responsibility of Controlling & protecting the data.
* The D.B.A should coridinate the design of the data base, guide the development & implimentation of data

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security processors, protect the integrity of data values & Make sure system performance is satisfactory.

Function of D.B.A :-

Database Design :-

* Conceptual database design consist of defining the data elements to be included in the database, the relationship is exist b/w them & the value constraints apply.
* A value constraints is a rule defining the acceptable values for a specific data items.
* Physical data base design determine the physical structure of the data base & include such decision as what access Methods will be used to retrieve data & what indexes will be built to improve the performance of the system.

User Training :- The D.B.A is responsible for educating users in the structure of the database & in its access through the D.B.M.S.

* This can be done informal training sessions by interacting with the users to create database views, through users manuals & periodic Memos & through Company information centers.

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Database Security + Integrity :- The concept of combining an organisation data into one common pool accessable to all has both advantages + disadvantages. Advantage of data sharing is offset by the disadvantage that data can be misused or damages that can be damaged by the users who do not have original responsibility & authority over the data.

* Access to of database is ultimately controled by a Password Mechanism where by a user attempting access gives a system validates passwords. The system allows the validated user only those access rights records in the data dictionary.
* Data Integrity refers to the problems of Maintaing the accuracy + the consistancy of data values.
* Security Mechanism such as password + data views protect data integrity.

Database System performance :- A database system being by simultaneously access by Many users may respond very slowly at times because the physical problems associated with users compating for the same resources are not trivial.

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Query Processor :→ Query processor helps the data base to simplify & facilitate to access the data. The Query processor components include,

* D.D.L (Data defination language) Interpreter :→

D.D.L Interpreter Interprets D.D.L statements & records the definition in the data dictionary.

* D.M.L Compiler :→ (Data Manuplation Language) :→

D.M.L Compiler translate D.M.L statements in a Query language into an evaluation plan consisting of low level instruction that Query evaluation engine understands.

* Query Evaluation Engine :→ It executes low level instruction generated by the D.M.L Compiler.

Storage Manager :→ Storage Manager is important because databases typically require a large amount of storage space.

* A storage Manager is a program Module that provides the interface b/w the low level data stored in the database & application programs & Queries submitted to the system.

* The storage Manager Component includes
  i) Authorisation & Integrity Manager.

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(i) Authorisation & Integrity Manager :-
Integrity Manager checks for the satisfaction of Integrity Constraints. & checks the Authority of users to access data.

(ii) Transaction Manager :- This ensures that the database remains in a consistent state even with the system failures & that concurrent transaction execution proceed without conflicting.

(iii) File Manager :- File Manager Manages the allocation of space on disk storage & the data structures use to represent information stored on disk.

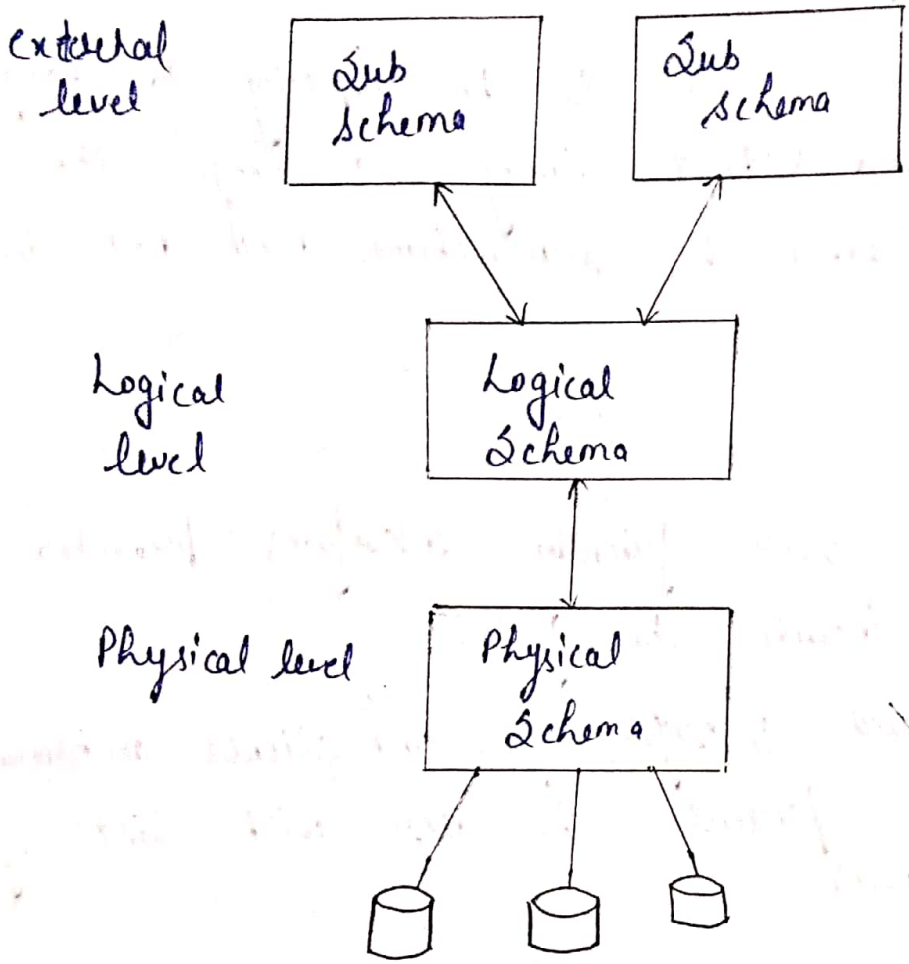
(iv) Buffer Manager :- This is responsible for fetching data from disk storage into Main Memory & deciding what data to be cache in Main Memory.

(v) Disk Storage :- The storage Manager implements several Data structure as part of the physical system implementation.

* Data files :- Data files store the data itself.
* Data dictionary :- This stores Metadata about the structure of the database.
* Indicies :- It can provide fast access to data item.

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Three Schema Architecture :→



Data Independence :→ The three schema architecture can be used to further explain the concept of data independence which can be defined as the capacity to change the schema at one level of data base system without having to change the schema at the next higher level, we can define two types of data independence.

(i) Physical data independence
(ii) Logical " "

* **The Logical data Independence :→** It is the capacity to change the conceptual schema without having to cha the external schema or application prooram. we may change the conceptual schema to expand the d

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by adding a record type or data item. (To reduce the database) (by removing a record type or data item). 12.

Physical data Independence :→ It is the capacity to change the physical schema without having to change the logical schema & hence the sub schema need not be change as well.

D.B.M.S Interfaces :→ User friendly interfaces provided by a D.B.M.S. May include the follow :-

* Menuplat Menubased Interface for web clients or browsing :-

*↳ These interfaces present the user with list of options. (Call Menus)

↳ Pulldown Menus are a very popular technique in web based user interface.

↳ They are also often used in browsing interface which allows a user to look through the content of a database in an unstructured manner.

* Formbased Interface :→

↳ It is place a form to each user.

↳ Users can fill out all of the form entries to insert new data all they can fill out only certain entries in which case the D.B.M.S. will retrieve Magic data for remaining entries.

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* Graphical User Interface :→ It Typically displace a schema to the user in diagramatic form.

↳ The user then can specify a Query by Manipulating the diagram.

↳ In Many cases graphical user interface utilizes both Menus and forms.

↳ Most graphical user interfaces use a pointing device such as a Mouse to pick certain parts of the displayed schema diagram

* Natural language Interface :→ These interfaces accept request & return in english or some other language & attempt to understand them

↳ A natural language interface usually has its own schema, which is similar to logical schema.

↳ The natural language interface refers to the words in its schema, as well as to the set of standard words in its dictionary to interpret the request.

* Speech input & output :→ Limited use of speech as an i/p Query & speech as an answer to a Question or result of a request is becoming commonplace

↳ The speech i/p is detected using a library of predefined words & used to set up the parameter that are supplied to the Queries.

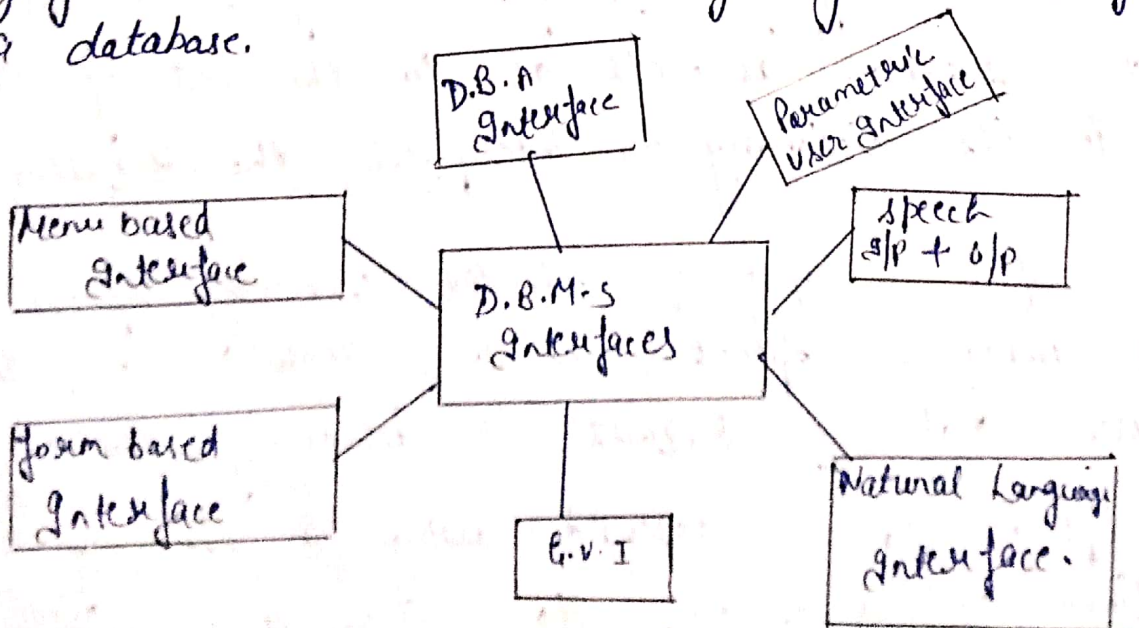
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↳ For o/p a similar conversion from text own no into speech takes place.

Interfaces for parametric users :→ Parametric users such as banktellers, often have a small set of operation that they must perform repeatidly. For ex :- teller is able to use single function keys to invoke routin + repeatative transections such as deposit or withdrawals into accounts, Balance inquaries.

Interfaces for the D.B.A :→ Most database system contain privileged commands that can be used only by the D.B.A staff.

↳ These includes commands for creating accounts setting system parameters, granting account authorisation, changing a schema + reorganising the storage structure of a database.



D.B.M.S Interface.

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E-R Model :→ It is a data Model which is based on the directly on the perception of real world. It is free from ambiguity & does not contain any technical information so it is easy to use even by non-technical person. The great advantage of E-R-diagrams is that they are easy to draw & understand. The E-R-Model employs three basic notation.

(i) Entity set
(ii) attributes
(iii) Relationship

1. Entity Set :→

* Entity :→ An Entity is any thing about which the System needs to store information for ex :→ Customers buy Products Employees sale products.

The nouns "Customers", "Products" & "Employees" are all Entities. It includes all those things about which can data can be collected. It can be of two types :-

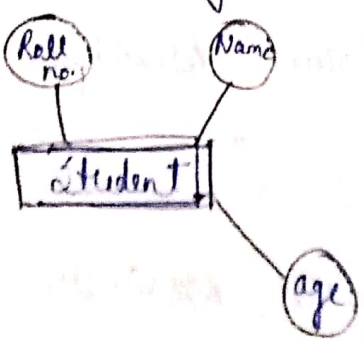
* Tangible
2. Intangible.

Tangible :→ Student, Pen etc. Means that exist physically.

Intangible :→ Bank account which exist only logically.

In relational table & entity is a tuple, row.

Ex →

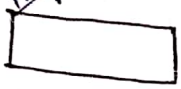


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Attributes :→ Attributes are the unique that describe the characteristics of entities a ER diagram. It is represented by eclipse.

↳ In Relational table it is represented by Columes.

ER Diagram Notation :→



→ entity Class



→ weak entity class.



→ Relationship Type.



→ Identifying Relationship Type



→ Attributes



→ Key Attributes



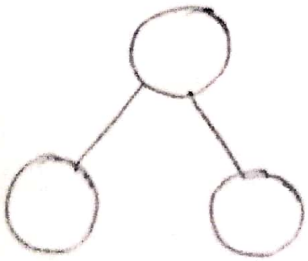
→ Partial Key Attributes



→ Derived Attributes



→ Multivalued Attribute



→ Composite Attribute.

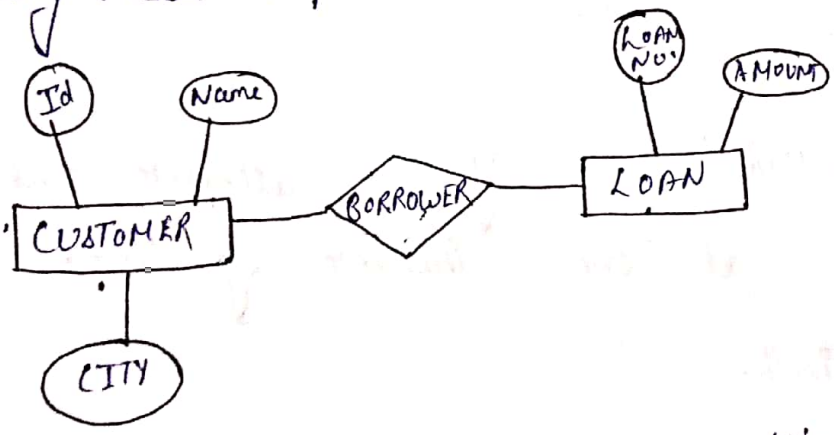
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Relationship Set :→ Relationship is the association b/w to or more entities in ER diagram. It is represented by diagonal form ◇.

Types of entities :→

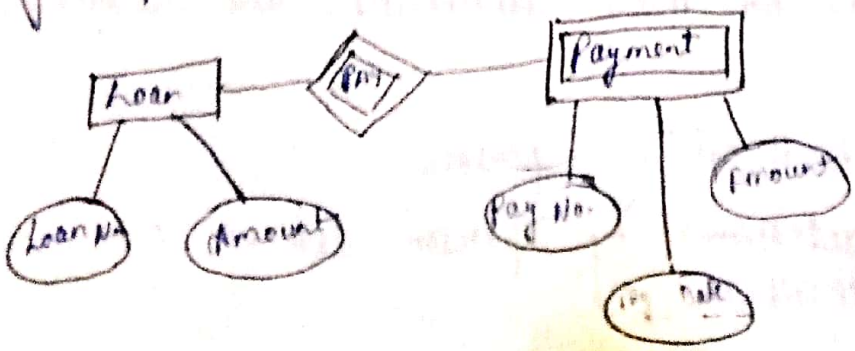
↳ **Strong Entity Set** :→ An entity set that has own primary key is called a strong entity set.

For ex:→ Consider an ER diagram which consist of two entity set, customer & loan related through a binary relationship



↳ **Weak entity Set** :→ A weak entity set does not have a primary key is called a weak entity set.

For ex:→ Consider an ER-diagram which consist of two entity set Loan & Payment related through a binary relationship in which entity payment is totally dependent on



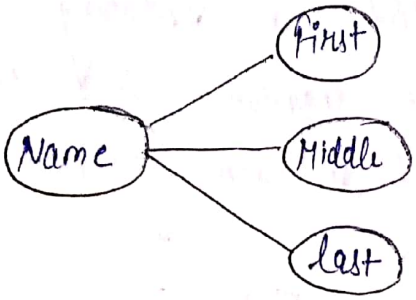
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Types of Attribute :-

* **Single value Attribute** :- If an attribute which are not divided into smaller sub parts are called Simple attributes

For ex :- I.D.

* **Composite Attributes** :- It can be divided into simple attributes.



* **Single Value Attribute** :- If an attribute contains only one value at one instance of time.

For ex :- Date of Birth.

* **Multivalue Attributes** :- which may have more than one value at any instant of time.

for ex :- Phone Number, e-mail I.D.

* **Derived Attribute** :- / **Stored Attribute** :-

Some times two or more attributes are related to each other.

for ex :- Age & birth date of a person.

Age can be determine by person date of birth So age is derived attribute.

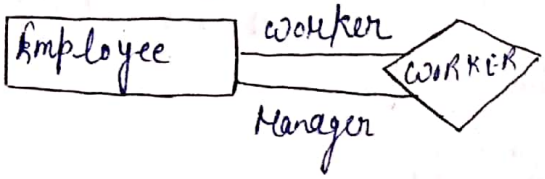
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↳ It is the data value which is actually stored in the data base.

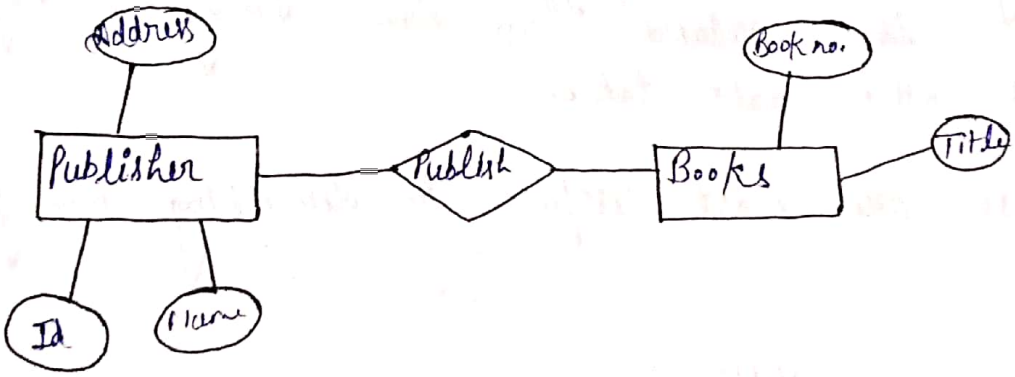
for ex → date of birth

Types of Relationship :-

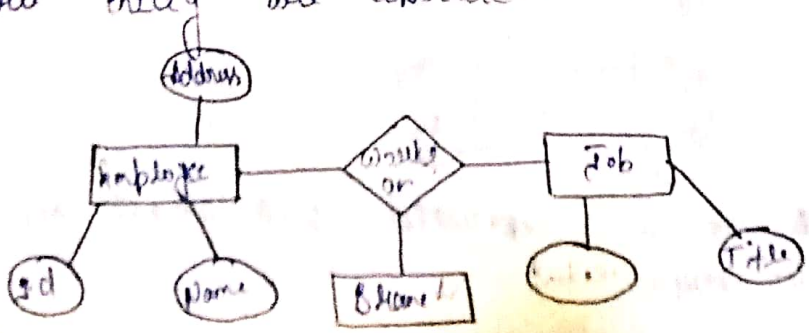
↳ Unary Relationship :- The unary relationship exist when an association is maintain with in a single entity



↳ Binary Relationship :- This relationship exist when two entities are associated.



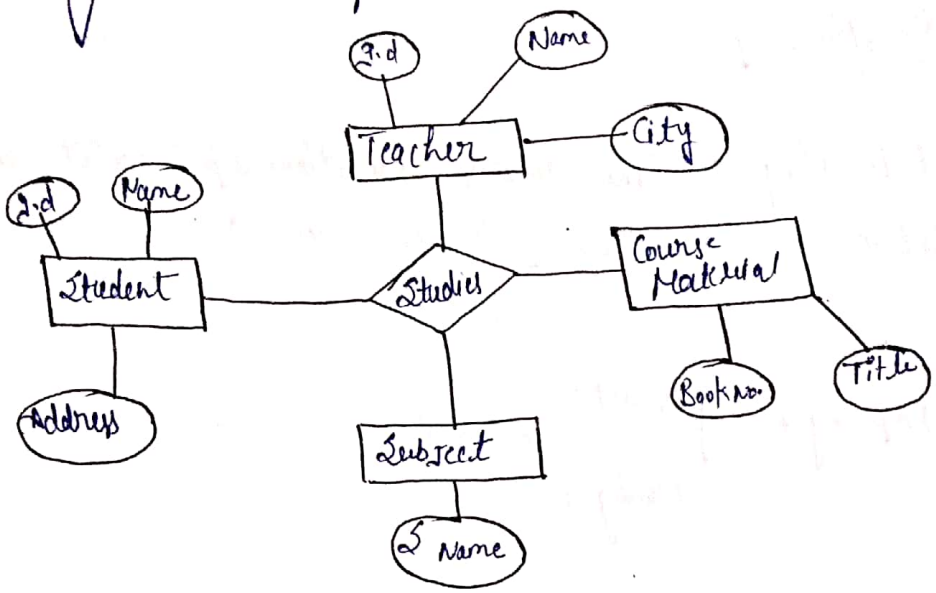
↳ Ternary Relationship :- This relationship exist when three entity are associated.



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Quaternary Relationship :-

This Relationship exist four entities are associated or Two Binary relationship are associated.

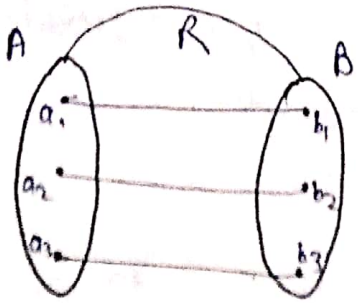


Mapping Cardinalities or Constraints :-

Mapping Cardinalities specify the no. of entities of the first table is related to how many no. of entities of the other ~~data~~ table.

* These are most useful in describing binary relationship sets.

↳ One to One Mapping :-

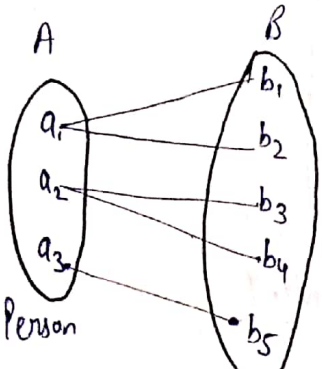


An entity in A is associated with almost one entity in B & vice-versa.

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For ex:- Student & Roll no.

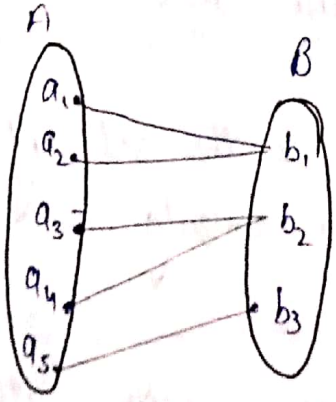
One to Many Mapping Relationship :- An entity in A is associated with any no. of B & entity in B is can be associated with atmost one entity in A.



Ex:- Person Mobile No. Mobile no.

1 : n

Many to one Mapping :- An entity in set A is associated with atmost one entity in set the B. An entity in set B can be associated with any no. of entities in set A.

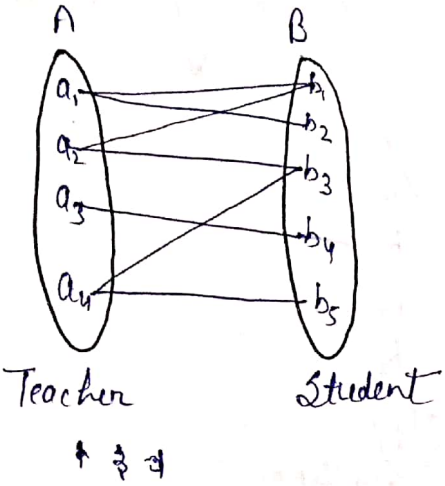


Mobile No. Person

n : 1

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↳ Many to Many Mapping :→ An entity in set A is associated with any no. of entity in set B. & An entity in set B with any no. of entities in set A.



Keys :→ A Key is an attribute ~~all~~ or a combination of attribute i.e. used to uniquly identify ~~reports~~ records. The purpose of the key is to bind data together across tables without repeating all of the data in every table. Different Types of keys are used in database.

↳ Super Key :→ Super key is the set of one or More attributes i.e. used to uniquly identify an entity In the entity set.

Ex:- Consider an entity employee with attributes { id, Name, address, salary }.

Super Key May be :-

{ Id }, { id, Name }, { id, Salary }, { Name, address }.

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Candidate Key :→ It can be defined as the minimal set of super key i.e. able to uniquely identify an entity in the entity set.

In other words candidate keys is the irreducable set of simple super key.

Ex→ Consider above example the candidate keys are I.d {I.d}, {Name, address}.

↳ Primary Key :→ A candidate key i.e. used by the data base designer to uniquely identify each entity in the entity set is known as primary key. A primary key can consist of one or more attributes of a table. Primary key is {Id}.

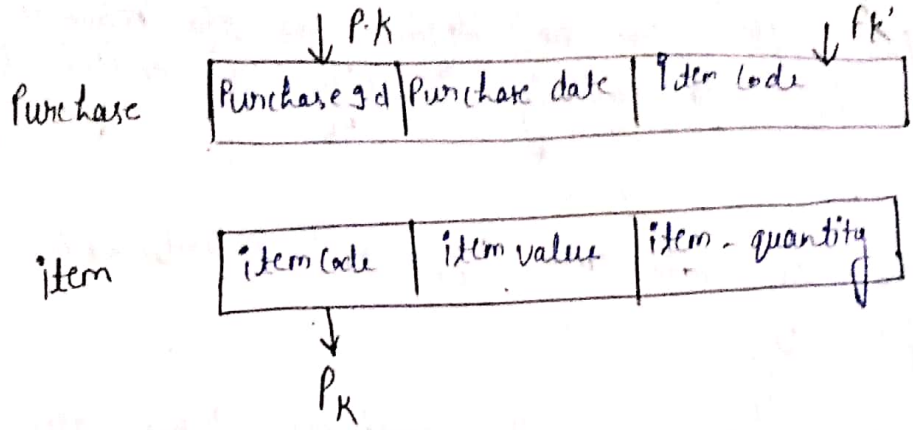
↳ Composite key :→ If we use multiple attributes to creat a primary key then that primary key is Composit Key. (Also called the compound key or concatinated key).

Ex→ Name, address.

↳ Foreign or Reference Key :→ A foreign is an attribute or combination of attribute in one base table that points to the primary key of another table.

Ex→ Consider to tables with fields.

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Enhance ER Model (EER) :→

* Specialization
* Generalization
* Aggregation

The ER Model which is supported to the additional semantic concepts is called the ~~the~~ enhanced ER Model. The E.ER Model includes all the concepts of the original ER Model together with the following additional concepts.

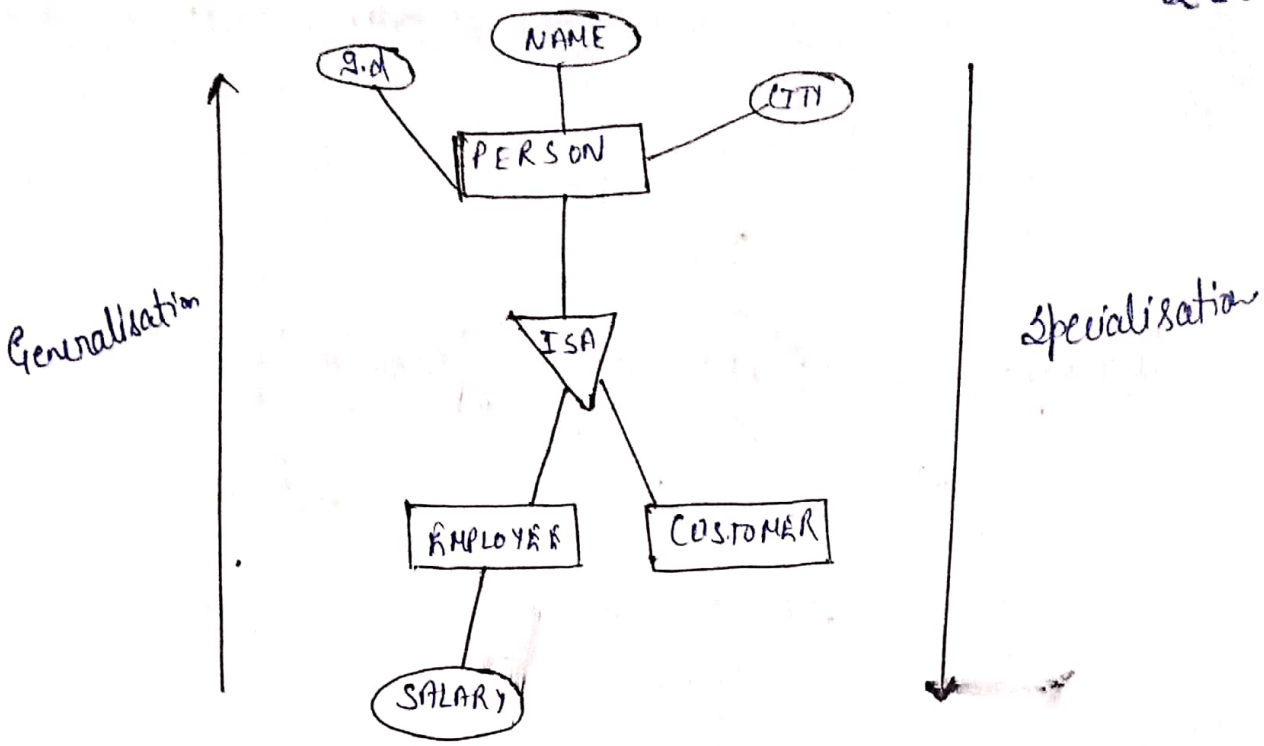
① **Specialisation** :→ Specialization is the process of designing sub groups with an entity set.

ex:→ Consider an entity set person with attributes id, name, address. A person may be further classified into Customer & employee

* In terms of ER diagram specialization is represented by a triangular component labeled ISA. ISA stands for "is a" & represent Customer is a person & employee is a person.

* The ISA relationship may also be refered as a super class, subclass relationship.
* It is a top of bottom approach.

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Specialisation & Generalisation

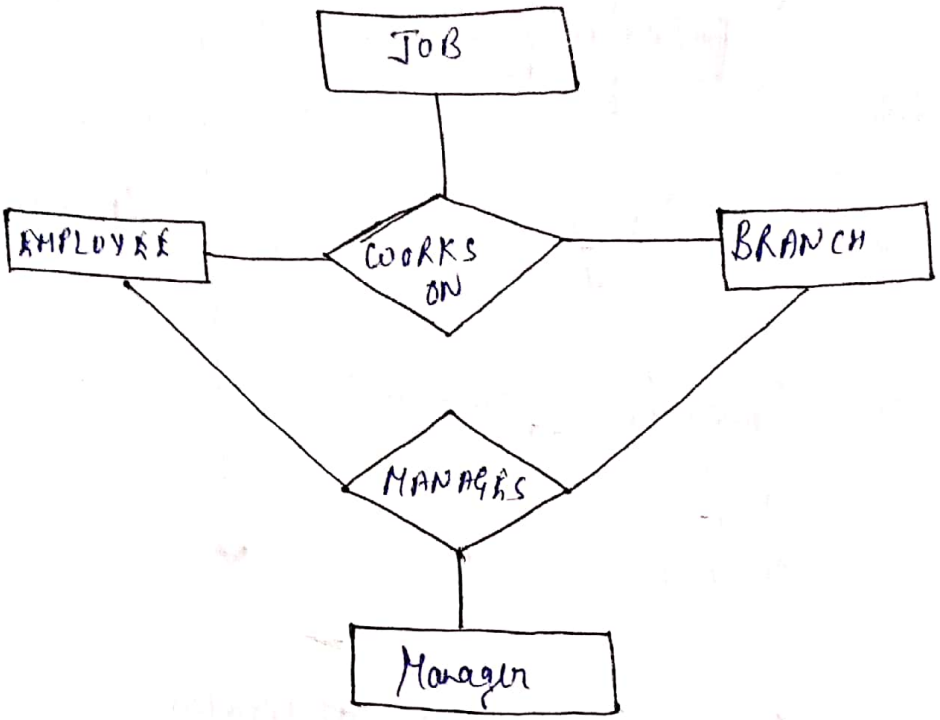
② **Generalization** :- It is a reverse process of Specialization.

↳ It is a bottom to Top approach in which multiple entities are synthesized into a higher level entity set on the basis of common feature.

↳ It is a process minimizing the difference b/w the entities by identify the common feature.

③ **Aggregation** :- One limitation of the ER Model is that it cannot express relationship among relationship. Consider an ER diagram.

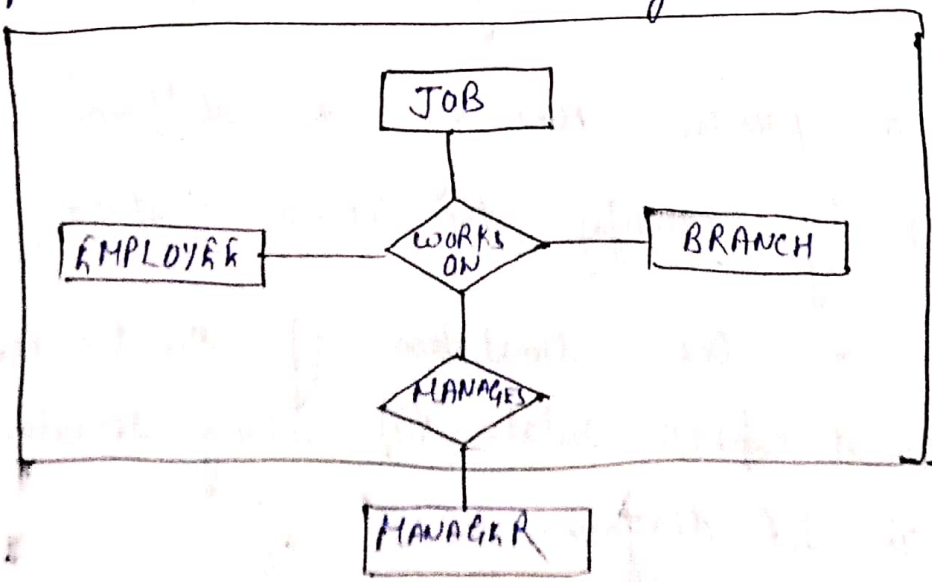
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(fig: ER diagram with redundant Relationship.)

There is redundant information in above figure the best way to model as situation such as the one just described is to use the aggregation

* Aggregation is an abstraction through which relationship are treated as higher level entities.



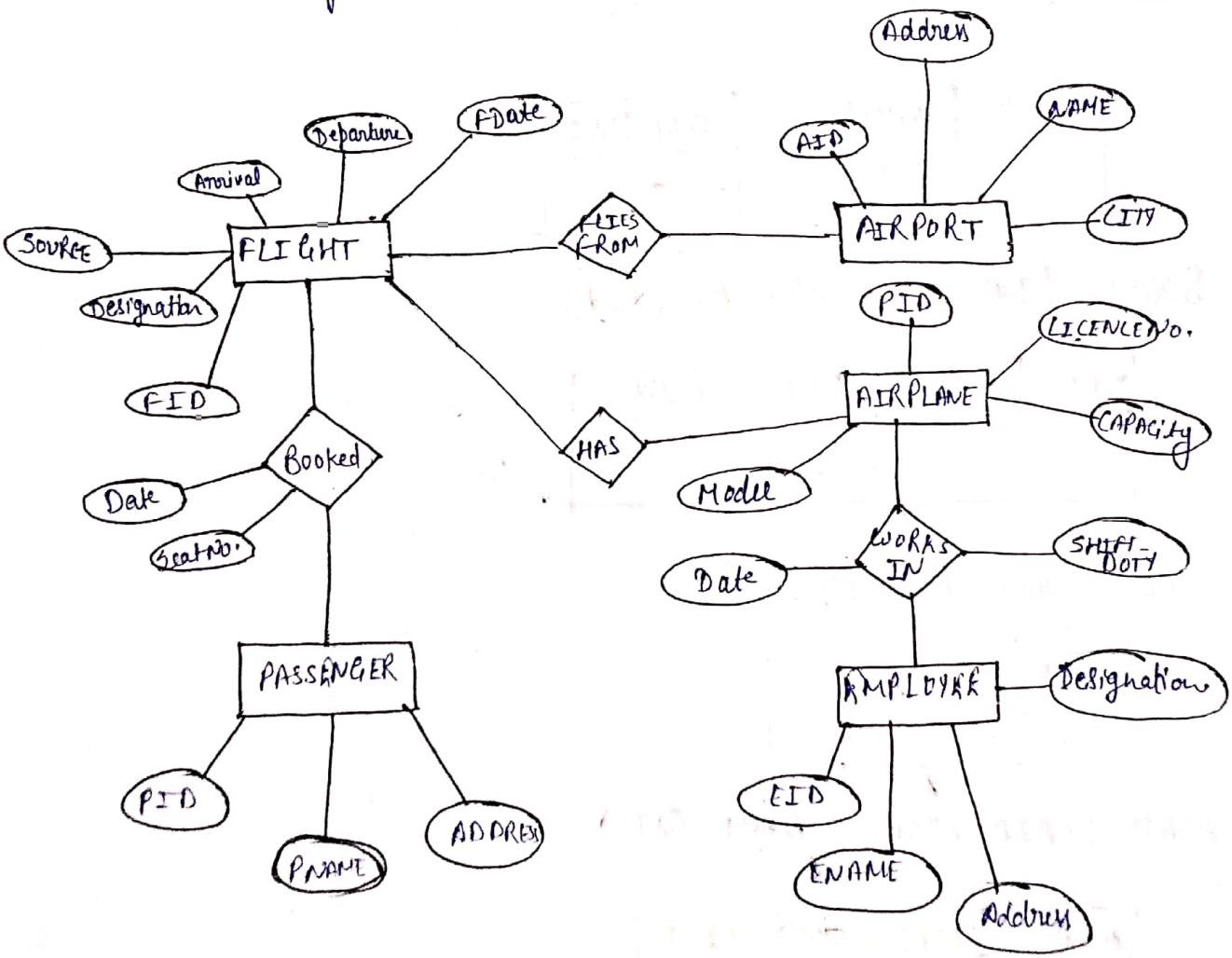
(fig:- ER diagram with aggregation)

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Reduction of an E-R Diagram to tables :->

1. Representation of strong entity sets.
2. Representation of weak entity sets.
3. Representation of relationship sets.

for ex :-> Consider an E-R diagram for airline reservation system



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Reduction of an E-R diagram into tables :→

FLIGHT (FID, SOURCE, DESTINATION, ARRIVAL, DEPARTURE, FDATE)

FID	SOURCE	DESTINATION	ARRIVAL	DEPARTURE	FDATE

AASSENGER (PID, NAME, ADDRESS)

PID	NAME	ADDRESS

BOOKED (DATE, SEAT No., FId, PId)

DATE	SEATNo.	FId	P.Id.

FLies from (FID, AID)

FID	AID

AIRPORT (AID, ADDRESS, NAME, CITY)

AID	ADDRESS	NAME	CITY

AIRPLANE (PID, MODEL, LICENCENO. CAPACITY)

PID	MODER	LICENCENo.	CAPACITY

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END

WORKSIN (DATE, SHIFT DUTY)

DATE	SHIFT DUTY

EMPLOYEE (EID, ENAME, Address, DESIGNATION)

EID	ENAME	ADDRESS	DESIGNATION

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